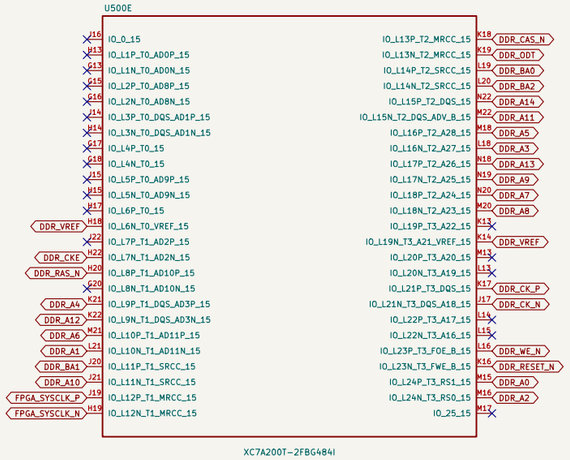
FPGA DDR3L memory and 200 MHz reference

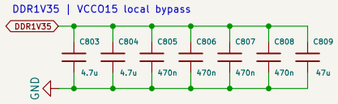
FPGA DDR3L address, command and clock

FPGA DDR3L address and command

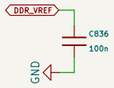
FPGA bank15 | command / clocks (E)



DDR1V35 | VCCO15 local bypass

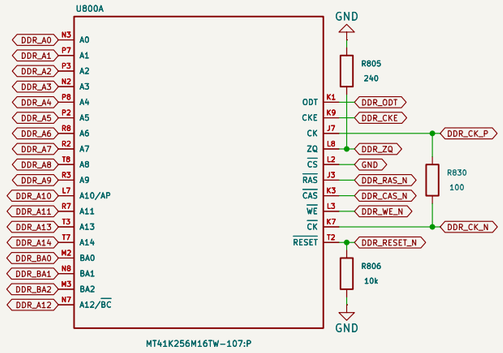


DDR_VREF | bank15 H18 / K14



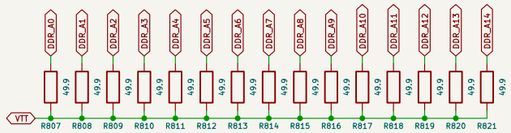
DDR3L address and command pins

DDR3L address / command (A)

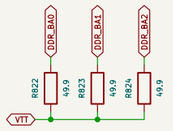


DDR3L signal termination

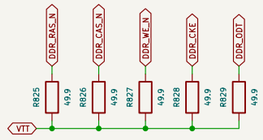
VTT | A0..A5



VTT | A12..A14 and BA0..BA2

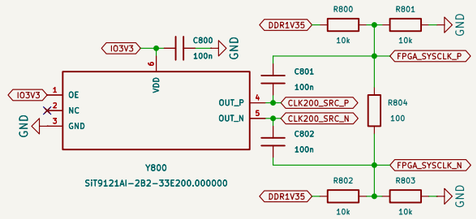


VTT | RAS / CAS / WE / CKE / ODT



FPGA 200MHz system clock

FPGA 200MHz reference | SYSCLK J19 / H19



Dedicated FPGA memory

U800 is an MT41K256M16TW-107:P DDR3L device, organised as 256M x 16 bits, or 512 MiB. FPGA bank 15 carries address, command and memory clock signals; bank 16 carries the two data byte lanes. This memory is independent of the MCU FMC SDRAM.
MIG target: 400 MHz CK / 800 MT/s, 4:1. CS# is grounded: disable Chip Select.

Address and data lanes

DDR_A0 through DDR_A14 and DDR_BA0 through DDR_BA2 address the memory. RAS_N, CAS_N, WE_N, CKE, ODT and RESET_N provide command and state control. DDR_DQ0 through DDR_DQ15 form the data bus. Each eight-bit lane has a differential DQS strobe and its own DM mask.

Termination and calibration

The 49.9 ohm resistor groups terminate the drawn address and control signals to VTT. R830 provides 100 ohms across DDR_CK_P/N. R805 is the 240 ohm ZQ calibration resistor to ground; R806 pulls DDR_RESET_N low. The memory chip-select pin is grounded in the shown single-device arrangement.

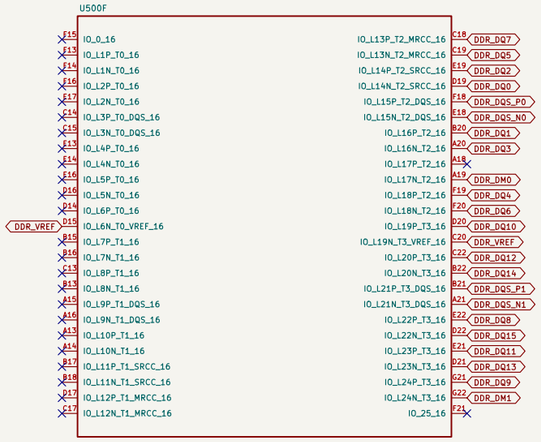
Memory power and references

DDR1V35 supplies memory VDD/VDDQ and FPGA VCCO15/VCCO16. DDR_VREF reaches both memory reference pins and the indicated bank reference balls. The dedicated 100 nF, 470 nF, 4.7 uF and 47 uF capacitor groups support these domains locally.

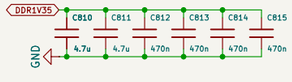
Reference clock and implementation

Y800 provides a 200 MHz differential reference. C801 and C802 AC-couple its outputs; four 10 kohm resistors bias the FPGA inputs from DDR1V35, and R804 terminates the pair with 100 ohms. FPGA memory-controller configuration and PCB routing must match the actual pin map, memory device and clock.

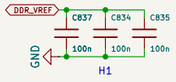
FPGA bank16 | data byte lanes (F)



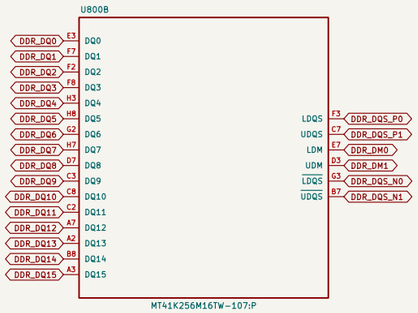
DDR1V35 | VCCO16 local bypass



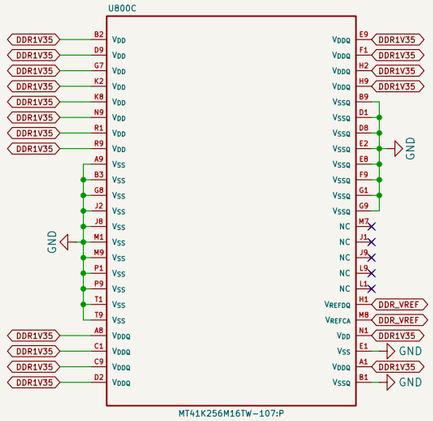
VREF bank16 | C20 / D15 M8 (A)



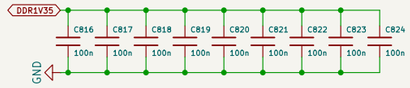
DDR3L DQ / DQS / DM (B)



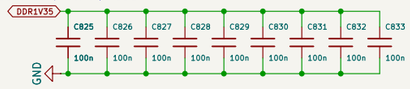
DDR3L supply / ground balls (C)



DDR1V35 | memory VDD



DDR1V35 | memory VDDQ



MIRA

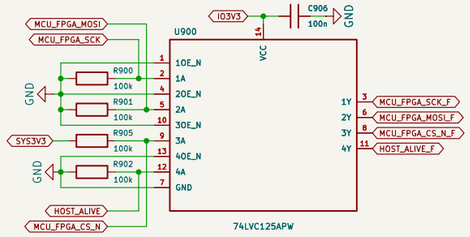
Sheet: /FPGA/800_FPGA_DDR3L_AND_REFERENCE_CLOCK/
File: 800_FPGA_DDR3L_AND_REFERENCE_CLOCK.kicad_sch

Title: FPGA DDR3L memory and 200 MHz reference

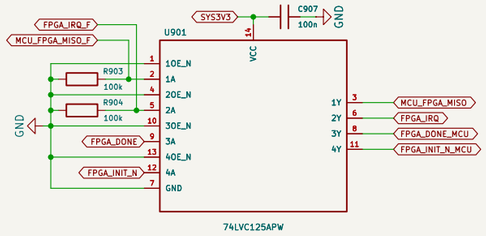
Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.8		Id: 12/43

Ethernet physical interface and MCU buffers

MCU to FPGA domain buffers

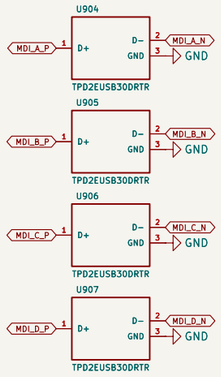


FPGA to MCU domain buffers



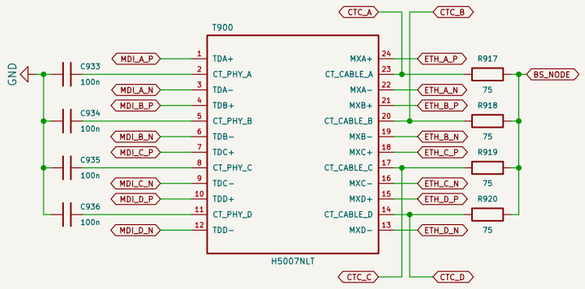
Ethernet RJ45, ESD and chassis

Ethernet RJ45, ESD and chassis | Rev.A

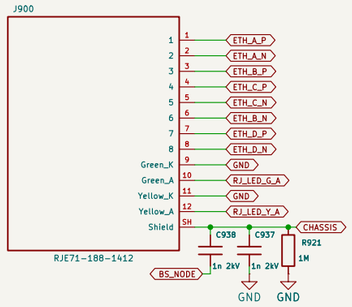


Ethernet magnetics and center taps

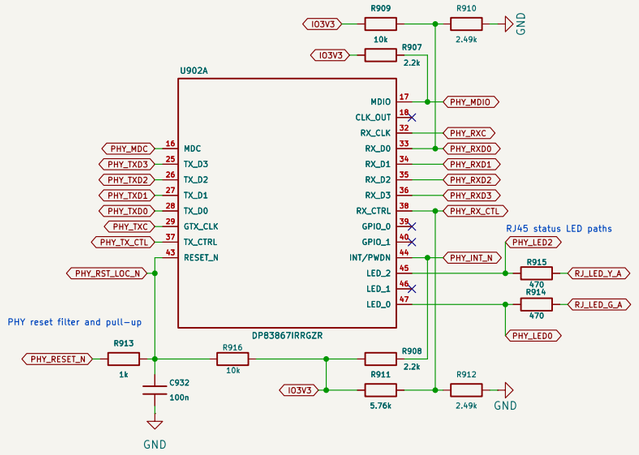
Ethernet magnetics and center taps | Rev.A



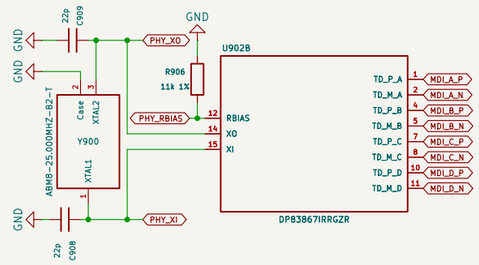
Ethernet RJ45, ESD and chassis

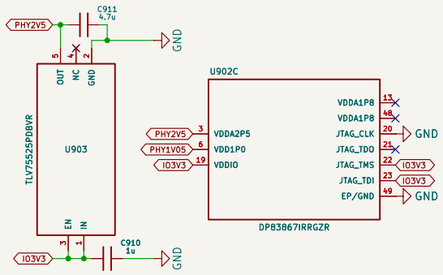


Ethernet RGMII and boot straps



Ethernet PHY supply and clock

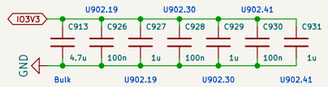




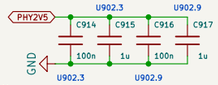
Ethernet RJ45, ESD and chassis

Ethernet PHY supply bypass capacitors | Rev.A

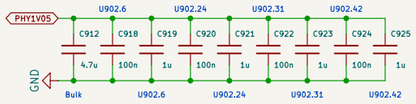
3.3 V I/O bypass



2.5 V analog supply bypass



1.05 V core supply bypass



Ethernet signal path

FPGA bank 13 connects by RGMII to U902, the DP83867IRRGZR PHY. Four MDI differential pairs pass through T900 magnetics to J900 RJ45. MDC/MDIO configure and read the PHY; PHY_INT_N reports events. Ethernet framing and network protocols are implemented by the FPGA design and software.

PHY clock, reset and straps

Y900 is a 25 MHz crystal with 22 pF capacitors. R906 is the 11 kohm bias resistor. Resistor pairs on PHY_RXD0 and PHY_RX_CTL establish startup strap levels. PHY_RESET_N reaches the local reset node through 1 kohm R913; R916 and C932 provide pull−up and filtering.

PHY supply domains

IO3V3 powers the I/O pins. U903 generates PHY2V5 from IO3V3 for the analog supply. PHY1V05 is a separate incoming core rail. The grouped 100 nF, 1 uF and 4.7 uF capacitors bypass the corresponding physical supply pins.

Cable interface and chassis

U904 through U907 protect the four Ethernet pairs. Four 75 ohm resistors join cable−side centre taps to BS_NODE, which couples to CHASSIS through C938. R921 and C937 connect chassis to circuit ground. R914 and R915 limit the RJ45 status LED currents.

Buffered MCU link

U900 forwards SCK, MOSI, CS_N and periodic HOST_ALIVE, powered by IO3V3. HOST_ALIVE is now a pulse train, not a static ready flag; FPGA must track it. U901 returns MISO, FPGA_IRQ, FPGA_DONE and FPGA_INIT_N from SYS3V3. Both use 74LVC125APW,118 with Ioff; C907 decouples U901. U901 3Y/4Y drive PC6/PC7; OE pins stay low, 100k define startup levels. SCK uses U500.Y11 (SRCC, bank 13); AB12 is unused. See digital−interfaces review.

MIRA

Sheet: /FPGA/900_FPGA_ETHERNET_AND_MCU_INTERFACE/
File: 900_FPGA_ETHERNET_AND_MCU_INTERFACE.kicad_sch

Title: Ethernet physical interface and MCU buffers

Size: A3	Date: 2026−10−18	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 13/43